

Name:

Student ID:

ELEC3115 Electromagnetic Engineering

T1, 2024

Midsession Test (Part A: Low frequency EM)

Total Marks: 50

Duration: 60 mins (includes 5 mins reading time)

It will contribute 15% toward your final marks.

You must return this question paper with your answer sheets.

1. Assumptions made in answering the questions should be stated explicitly.
 2. All answers must be written in ink. Pencils may only be used for drawing, sketching or graphical work.
 3. Useful constants and formula sheet are provided at the back of the question paper.
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Question 1 [10 marks]

- (a) [2+3 =5 marks] Restate in your own words the first postulate of electrostatic and show how it becomes Gauss's law in its integral form.
- (b) [5 marks] A capacitor consists of two parallel plates of cross-sectional area 250 cm^2 . The plates contain equal but opposite charges of value $\pm 5 \text{ } \mu\text{C}$ and are separated from each other by 1 mm . The dielectric material between the plates is air.
- (i) [2 marks] Under what assumption, can you use Gauss's law to calculate the electric flux density vector $\vec{D}$ between the two plates?
- (ii) [2 marks] Calculate the electric flux density vector $\vec{D}$ between the two plates.
- (iii) [1marks] What is the value of flux density when separation distance 1 mm is reduced to 0.5 mm ?

Question 2

(a) [15 marks]

- (i) [4 marks] A very long transmission line is located above the ground at a height $h = 8 \text{ m}$. You may assume influence of ground plane is negligible at this height. The line is charged

with a uniform line charge density $\rho_l = 100 \text{ nC/m}$. If the electric field at any arbitrary radial distance r from the centre of the line is $\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \vec{a}_r$, show that potential difference at a radial distance r_2 with respect to a reference at radial distance r_1 from the centre of the transmission line is given as, $V = \frac{\rho_l}{2\pi\epsilon_0} \ln\left(\frac{r_1}{r_2}\right)$.

(ii) [2 marks] If the diameter of the transmission line is 1.2 cm, what is the potential difference at the surface of the line with respect to a reference located at the ground below?

(iii) [3+1 =4 marks] Calculate the capacitance per meter between the line and the ground using Method of Image. Why do we need to use 'Method of Image' for this calculation?

(b) [5 marks] Capacitance formation in a 3-conductor transmission system is shown in Figure 1. V_1 , V_2 , and V_3 are the potential of the conductors. Capacitance values obtained from the measurement are shown in the figure. Find the capacitance matrix of this network.

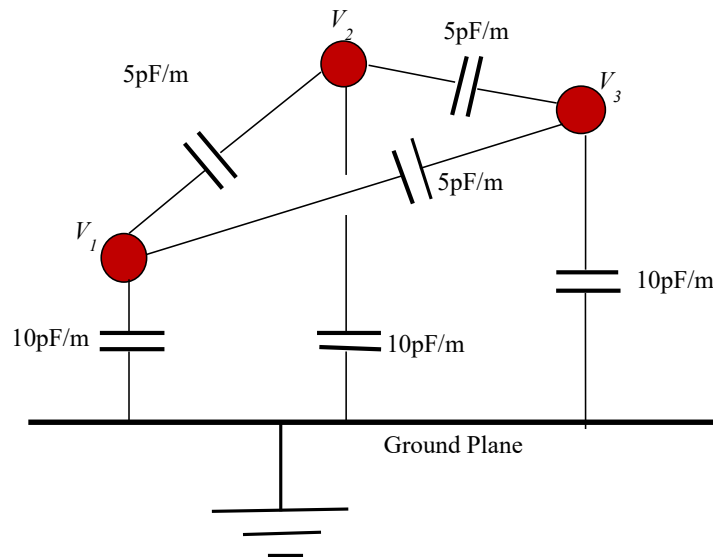


Figure 1

Question 3 [15 marks]

(a) [2 marks] What is the definition of dielectric strength of an insulating material?

(b) [12 marks] The top view of a coaxial cable is shown in Figure 2. The radius of the inner conductor $a = 2 \text{ mm}$, and the thickness of the metallic shield is 4 mm. An insulating

material of permittivity $\epsilon=3\epsilon_0$ F/m fills the space between the inner conductor and the outer shield. The dielectric strength of the material is 25 MV/m.

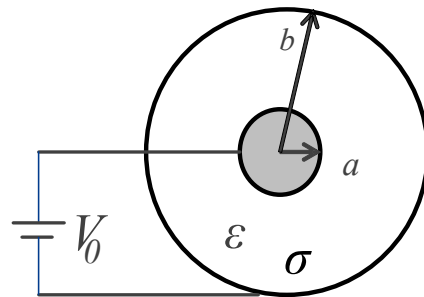


Figure 2

- (i) [5 marks] Calculate the maximum voltage V_0 that can be applied between the inner conductor and the outer shield without causing a breakdown of the insulating material.
- (ii) [3 marks] Calculate the capacitance per meter of the cable.
- (iii) [3 marks] The insulating material of the cable is an imperfect dielectric and hence, it also has a conductivity $\sigma=5.13 \times 10^{-9}$ S/m. Calculate the associated leakage resistance between the inner conductor and the outer shield.
- (iv) [2 marks] What is the implication when a leakage resistance is present in a coaxial cable?

Question 4 [10 Marks]

- (a) [2 marks] What is relative permeability, and why do we use it instead of a magnetization vector?
- (b) [4 marks] The mean length of the steel core of a toroidal inductor is 1 m, the area of cross-section of the core is 7 cm^2 . The relative permeability of the core steel is 5000. If the applied MMF is 25 At, calculate the flux in the core. Assume uniform flux density.
- (c) [2+2 = 4 marks] Does a small air gap of a few mm introduced in the toroid core affect the core flux? Why or why not? Is an air gap in the core of an toroidal inductor desirable? Why or why not?

End of the paper

(See the next page for the formula sheet)

Useful formula and constants:

$$\varepsilon_0 = 8.854 \times 10^{-12} \text{ C/m}$$

$$\vec{\nabla} \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\vec{\nabla} \times \vec{A} = \vec{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \vec{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \vec{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\oint_s \vec{E} \cdot d\vec{s} = \frac{Q_{enclosed}}{\varepsilon} \quad \text{or} \quad \oint_s \vec{D} \cdot d\vec{s} = Q_{enclosed}$$

$$Q_{enclosed} = \int_v \rho dv \quad \text{or} \quad \int_s \rho_s ds \quad \text{or} \quad \int_l \rho_l dl$$

$$V_{ba} = - \int_a^b \vec{E} \cdot d\vec{l}$$

$$\vec{P} = \vec{D} - \varepsilon_0 \vec{E}, \quad \vec{D} = \varepsilon \vec{E}, \quad \varepsilon = \varepsilon_0 \varepsilon_r$$

$$\vec{E} = \left| \frac{\rho_l}{2\pi\varepsilon r} \right| \vec{a}_r, \quad V/m, \quad V_{21} = \frac{\rho_l}{2\pi\varepsilon} \ln \left(\frac{r_1}{r_2} \right)$$

$$|E_{1t}| = |E_{2t}|, \quad |D_{1n}| = |D_{2n}|$$

$$C = \frac{\pi\varepsilon}{\ln \frac{d-a}{a}} \quad F/m, \quad C = \frac{2\pi\varepsilon}{\ln \left(\frac{b}{a} \right)} \quad F/m$$

$$w_s = \frac{1}{2} \vec{D} \cdot \vec{E} \quad [J/m^3], \quad \vec{F}_s = \pm \vec{\nabla} W_s$$

$$\vec{\nabla}^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \varphi^2} + \frac{\partial V^2}{\partial z^2} = 0$$

$$C = \frac{\pi\varepsilon_0}{\ln \left[\left(\frac{d}{2a} \right) + \sqrt{\left(\frac{d}{2a} \right)^2 - 1} \right]} \quad F/m$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\oint_c \vec{B} \cdot d\vec{l} = \mu_0 I_c, \quad \oint_c \vec{H} \cdot d\vec{l} = I_{enclosed}$$

$$\vec{B} = \vec{\nabla} \times \vec{A},$$

$$\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{M}, \quad \vec{B} = \mu_0 (1 + \chi_m) \vec{H} = \mu_0 \mu_r \vec{H} = \mu \vec{H}$$

$$\vec{\nabla}^2 \vec{A} = -\mu_0 \vec{J}$$

$$\Phi = \int_s \vec{B} \cdot d\vec{s}, \quad \Re = \frac{l}{\mu S},$$

$$L = \frac{\lambda = N\phi}{i},$$

$$EMF = -N \frac{d\phi}{dt}, \quad EMF = \oint_c (\vec{u} \times \vec{B}) \cdot d\vec{l}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$V_{rms} = 4.44 fNB_m S_c$$

$$\delta = \sqrt{\frac{2}{\sigma\mu\omega}}, \quad R_{ac} = \frac{1}{\delta\sigma} \left(\frac{l}{\pi(D-\delta)} \right)$$

$$P_h = k_2 B_m^n f, \quad P_e = k_1 B_m^2 f^2$$

$$\vec{J} = \rho \vec{u}, \quad I = \int_s \vec{J} \cdot d\vec{s}$$

$$\vec{J} = \sigma \vec{E}, \quad R = \frac{l}{\sigma S}, \quad P = \int_v (\vec{E} \cdot \vec{J}) dv'$$

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}, \quad \rho = \rho_0 e^{-\left(\frac{\sigma}{\varepsilon}\right)t}$$

$$\vec{\nabla} \times \vec{J} = 0$$

$$RC = \frac{\varepsilon}{\sigma}$$